

MARKING SCHEME
THE ABERDARE TECHNICAL COLLEGE
ICT TECHNICIAN LEVEL 5
CAT 2 term 2

1. Switch vs Router (2 Marks)

- Switch operates at **Layer 2 (Data Link)** and connects devices within a LAN (1 mark)
- Router operates at **Layer 3 (Network)** and connects different networks / routes packets (1 mark)

2. Transmission media considerations (2 Marks)

Any two:

- Distance of transmission (long/short distance suitability)
- Speed/bandwidth requirements
- Cost of installation
- Environment (interference, EMI)
- Security requirements

(1 mark each)

3. Fibre optic vs copper cable differences (2 Marks)

Any two:

- Fibre uses light signals, copper uses electrical signals (1)
- Fibre has higher speed/bandwidth (1)
- Fibre is immune to EMI, copper is affected (1)
- Fibre supports longer distances (1)

(Max 2 marks)

4. Access point role (2 Marks)

- Provides wireless connectivity to devices (1)
- Bridges wireless devices to wired LAN (1)

5. Network security devices (2 Marks)

Any two devices with function:

- Firewall – filters incoming/outgoing traffic
- IDS/IPS – detects/prevents intrusion
- Proxy server – controls access and filters content

(1 mark device, 1 mark function)

6. Networking software vs management tools (2 Marks)

- Networking software: OS/software that enables network operations (e.g., Windows Server, Linux) (1)
- Management tools: tools used to monitor/manage network performance (e.g., Wireshark, PRTG) (1)

7. Server functions (2 Marks)

Any two:

- Provides resources/services (files, printers)
- Manages client requests
- Centralized data storage
- Authentication/authorization

(1 mark each)

8. Storage systems uses (2 Marks)

Any two:

- Backup of network data
- Central file storage
- Data sharing among users
- Disaster recovery

9. Multimeter purpose (2 Marks)

- Measures voltage, continuity, resistance in cables (1)
- Used to troubleshoot faulty network/electrical connections (1)

10. Patch cord vs backbone cable (2 Marks)

- Patch cord: short cable for device-to-device connection (1)
- Backbone cable: main cable carrying network traffic between segments (1)

11. Tools functions and safety (10 Marks total)

a) Crimpers

- Function: attaches RJ45 connectors to cables (1)
- Safety: ensure correct pressing to avoid injury / damage (1)

b) Strippers

- Function: removes cable insulation (1)
- Safety: avoid cutting inner wires (1)

c) Punch down tool

- Function: terminates cables in patch panels/sockets (1)

- Safety: hold firmly to avoid slipping (1)

d) Network tester

- Function: tests cable continuity and faults (1)
- Safety: ensure correct connection before testing (1)

e) Tone generator/probe

- Function: traces cables in installation (1)
- Safety: avoid use on live high-voltage circuits (1)

12. Safety precautions (3 Marks)

Any three:

- Switch off power before installation
 - Use insulated tools
 - Handle tools carefully
 - Avoid wet environments
 - Wear PPE
- (1 mark each)

13. Fiber optic tools applications (2 Marks)

Any two:

- Stripping fiber cables
- Cleaving fiber ends
- Splicing fibers
- Testing optical signal loss

(1 mark each)

14. Network materials functions (10 Marks)

- a) Network cables – transmit data signals (2)
- b) RJ45 sockets – provide network connection points (2)
- c) Patch cords – connect devices to network outlets/switches (2)
- d) Trunking covers – protect and organize cables (2)
- e) Cable ties – bundle and secure cables (2)

15. LAN components (4 Marks)

Any four:

- Switch
- Router
- Computers
- Access point
- Server
- Cables (UTP/fibre)
- Patch panel

(1 mark each)

16. Network component considerations (3 Marks)

Any three:

- Compatibility with existing systems
- Scalability (future expansion)
- Cost
- Performance requirements
- Environment suitability

(1 mark each)

17. LAN setup procedure (4 Marks)

Any correct steps:

- Plan network layout (1)
- Install cabling and trunking (1)
- Connect devices to switch/router (1)
- Configure IP addressing and test connectivity (1)

18. Challenges and solutions (4 Marks)

Any two:

1. Cable faults → use network tester and re-terminate cables (2)
2. IP conflicts → configure DHCP or correct IP settings (2)
3. Interference → use proper cable type/shielding (2)
4. Misconfiguration → reconfigure network devices (2)

(2 marks each challenge + solution)

19. Tools and uses (2 Marks)

Any two:

- Crimping tool – attaches RJ45 connectors (1)
- Network tester – tests cable connectivity (1)
- Punch down tool – terminates cables (1)
- Multimeter – checks continuity/voltage (1)